Uwagi

2: KS: tytuł
9: ZL: Nie wiem, czy opisywać pozostałe operacje logiczne, tylko te dwie są wykorzystywane
9: KS: Na razie nie :-)
9: KS: `https://github.com/ZofiaLor/magister-granule`
12: KS: ¿Czy tutaj są dobre wartości?
18: ZL: To jest bardziej skomplikowany sposób, który był omawiany w pracy magisterskiej. Poza tym, że mogły pojawić się szumy, nie wprowadził on żadnych nowych informacji. Wszystkie tabele z wartościami wyników, które zostały w artykule, dotyczą prostszej metody, opisanej w następującym akapicie:
18: KS: To oznaczenie trzeba będzie zmienić na coś sensowniejszego, ale na razie to tak zostawiam.
19: ZL: Do wyrzucenia, jeśli nie opisujemy metody z alg. 2
19: ZL: Zależność odwróconą mamy też dla 'blobs' – może warto jednak pokazać ten wykres?
19: KS: Tak, dajmy to.
20: KS: Czy to jest związane z tym, że trójkątna relacja może generować szum?
20: ZL: Tabela? Wykres?

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GrAgglo: A granular agglomerative clustering algorithm [tytuł]

5th November 2026

Abstract

Keywords: hierarchical clustering, fuzzy C-means, granular computing, fuzzy numbers, Python

1 Introduction

Clustering is a very common technique of the unsupervised data analysis. It has been widely used in many applications. Many techniques and algorithms have been proposed, implemented, and used.

The idea of data clustering is the partition of data into groups. Items in the same group are more similar to each other than to those in other groups. The similarity is usually defined by a distance function. Commonly the Euclidean distance is used, but other distance functions can be applied as well.

Multiple clustering techniques have been proposed in the literature. One of them is the hierarchical clustering. In the ascending hierarchical clustering, each data item is initially assigned to a separate cluster. Then, the closest clusters are merged into one cluster. This technique is not complicated, however it has a high time complexity, which is in general $O(n^3)$, where n is the number of data items. This may be a bottleneck in enormous data sets.

This problem can be addressed with granular computing. Granular computing (GrC) is a paradigm of data analysis with informative granules. The data are gathered in granules and the granules are an input of further algorithms. The representation of granules heavily depends on the application. Commonly set, fuzzy sets, rough sets, intervals and many other are used for granules.

Unfortunately, the mathematical representation of granules is not straightfoward for mathematical operations. Commonly, Zadeh's extension principle is used to elaborate the results. It causes three main problems [29]:

- for some fuzzy sets, interative numerical procedures are needed, the operations cannot be presented with closed-form formulae;
- for some fuzzy sets and operations, the fuzzy set in the result of an operation does not preserve the properties of the operands (the set of fuzzy numbers with operations is not complete);
- multiple arithmetical operations causes spread of fuzziness, the more operations run, the more fuzzy result, even beyond reasonable interpretability.

In this paper, we would like to address these two problems. We would like to use granular approach to accelerate hierarchical clustering algorithms with extensional fuzzy numbers.

2 Related works

2.1 Hierarchical clustering algorithms

Clustering is a well known technique for unsupervised data analysis. It aims at elaboration of a partition of data item in such a way that items in one cluster are more similar to items in the same cluster than to items in other clusters [8, 41].

Multiple paradigms are used for clustering algorithms [41], eg.:

- Hierarchical approach based on either merging small clusters into larger ones (ascending approach) or splitting larger clusters into smaller ones (descending approach) [41], eg. single-linkage [7, 32], SLINK [21], complete-linkage [34], gravitational clustering [39];
- Criterion function approach based on minimalisation of a certain criterion is represented by a huge FCM [2], family, eg. FCOM [12], FuBi [27], Rough FCM [24, 22], Subspace FCM [23] etc.
- Density approach locates clusters in the most dense regions of the input domain, eg. DBSCAN [6, 35], OPTICS [1].
- Graph approach treats data items as graph vertices and similarities as graph edges, eg. based on spectral clustering [15], deep methods [20], kernels [13], density [36], modularity [3], probability [37].

Hierarchical clustering algorithms are based on the idea of merging or splitting clusters in a hierarchical way. The ascending hierarchical clustering algorithm

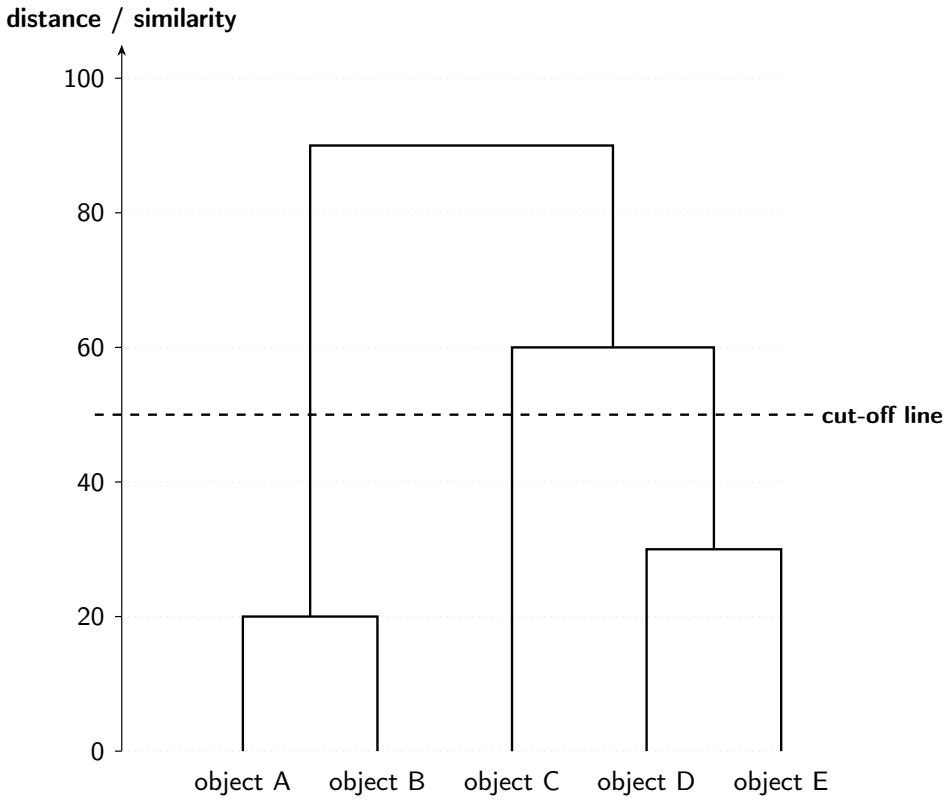


Figure 1: Example of a dendrogram for hierarchical clustering of five objects with a cut-off line at distance 50.

starts with each data item in a separate cluster. Then, the closest clusters are merged into one cluster. This process is repeated until all data items are in one cluster. Clustering can be visualized with a dendrogram (Fig. 1), which is a tree-like diagram that shows the hierarchical relationships between clusters. The height of the branches in the dendrogram represents the distance between clusters. A cut-off line can be drawn to determine the number of clusters. The distance between clusters can be defined in different ways, such as single-linkage, complete-linkage, average-linkage, or Ward's method. This choice affects the shape and size of the resulting clusters (Fig. 2). In the single-linkage approach [7, 32], the distance d between two clusters is defined as the minimum distance between any pair of points in the two clusters $\mathbb{A}$ and $\mathbb{B}$:

$$t\left(\mathbb{A}, \mathbb{B}\right) = \min_{\mathbf{a}\in\mathbb{A}, \mathbf{b}\in\mathbb{B}} t\left(\mathbf{a}, \mathbf{b}\right). \tag{1}$$

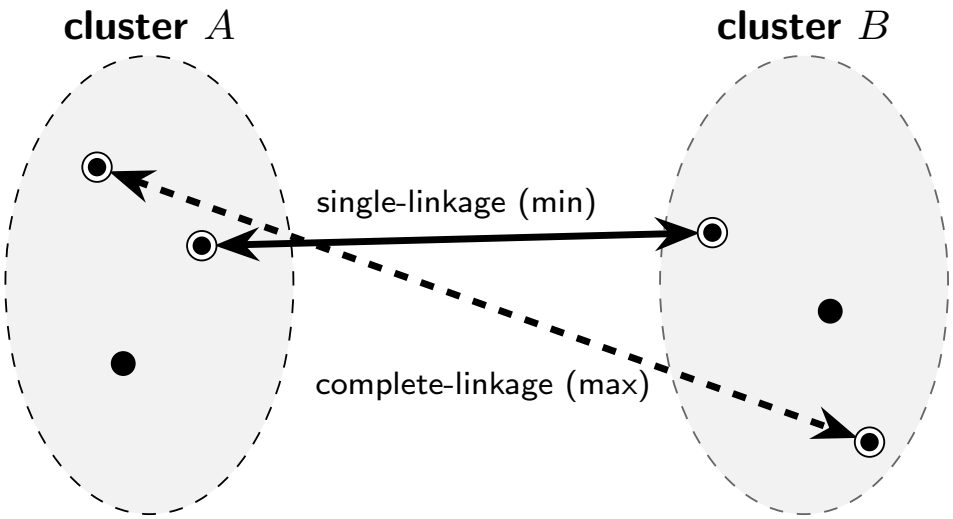


Figure 2: Example of single-linkage and complete-linkage methods for two clusters.

For complete-linkage [34, 14, 5], the distance between two clusters is defined as the maximum distance between any pair of points in the two clusters:

$$t\left(\mathbb{A},\mathbb{B}\right)=\max_{\mathbf{a}\in\mathbb{A},\mathbf{b}\in\mathbb{B}}t\left(\mathbf{a},\mathbf{b}\right). \tag{2}$$

In the complete-linkage approach when the distance between all data items do not exist, the dendrogram is not connected and has several not connected components.

Some more strategies are also used for defining the distance between clusters, such as WPGMA (Weighted Pair Group Method with Arithmetic Mean) [33], UPGMA (Unweighted Pair Group Method with Arithmetic Mean) [33], or the Ward's method [38].

This approach has a high time complexity. It is necessary to compute the distance between all pairs of clusters in each iteration, which results overall time complexity of $O(n^3)$. There are known implementations of single-linkage SLINK [21] and complete-linkage CLINK [5] algorithms with time complexity of $O(n^2)$. However, there are some limitations of these algorithms, eg. SLINK cannot handle a case when more pairs of clusters have the same minimal distance. In contrast, the FCM algorithm has a time complexity of $O(n)$, which is significantly lower.

2.2 Granular computing

Lofti Zadeh introduced a new paradigm of computation based on informative granules [46]. This is an abstraction that fills the gap between numbers and human concepts. Humans commonly do not operate on numbers in every day life. We simple operate on concepts that can be described with numbers. Zadeh named these concepts informative granules.

The main idea of granular computing (GrC) is to provide ideas, methods, algorithms, and techniques to make computer work in a way that is similar to human cognition [47]. The next step dreamt by Zadeh were computers calculating with words [48, 17].

Informative granule is a collection of similar items with a clear significance. This definition is very general. It does not define the similarity of the items nor the significance. These notions are specified in an application they are used for. Thus the representation of granules is very diverse [29], eg. intervals, sets, fuzzy sets, interval type-2 fuzzy sets [16], rough sets [31, 18], intuitionistic sets, shadowed sets, clusters [25], soft sets [40], if-then rules [26], extensional fuzzy numbers [29].

The next level of abstraction over informative granules is the triarchic theory of GrC based on three perspectives mutually intertwined: philosophical, methodological, and computational [44]. This is a base of the three-way decision approach [42, 43, 28, 19, 30].

One of representations of informative granules are extensional fuzzy numbers.

2.2.1 Extensional fuzzy numbers

Extensional fuzzy number is a number constructed with a extension of crisp numbers with a fuzzy $\star$-similarity relation defined for a left continuous t-norm $\star$ [9, 10, 11, 29]. This is an alternative to the Zadeh's extension principle [45] way for constructing fuzzy numbers. It has some advantages such as [29]:

- Extensional fuzzy numbers are complete with respect to the arithmetic operations: the result of an operation on extensional fuzzy numbers is a extensional fuzzy number.
- The infinite support of a fuzzy number is allowed.
- There is no spread of fuzziness: interative operations on extensional fuzzy numbers do not increase the vagueness of the result.
- The computational overload of extensional fuzzy numbers is very low: no discretisation, no integration, no iterations are needed.

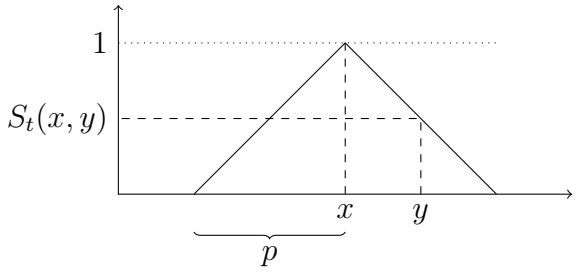


Figure 3: Fuzzy $\star$-similarity relation S_t [29].

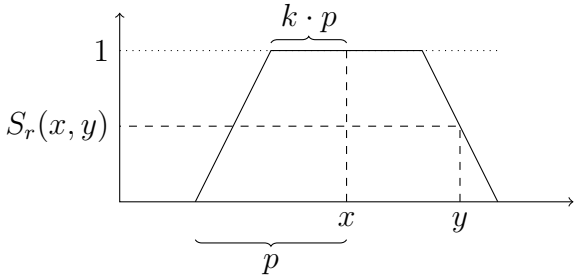


Figure 4: Fuzzy $\star$-similarity relation S_r [29].

- Infinite number of types of extensional fuzzy numbers can be easily constructed with various fuzzy $\star$-similarity relations.

Extensional fuzzy numbers can be constructed with any fuzzy $\star$-similarity relation that is defined as a left continuous t-norm that is reflexive, symmetric and satisfies the triangle property [29]. $\star$-similarity relations that correspond with common fuzzy sets are [29]:

- for symmetric triangular fuzzy sets (Fig. 3)

$$S_t(x, y) = \max\left\{1 - \frac{|x - y|}{p}, 0\right\}, \tag{3}$$

  where $p > 0$;

- for symmetric trapezoidal fuzzy sets (Fig. 4)

$$S_r(x, y) = \min\left\{1, \max\left\{1 - \frac{|x - y|}{p}, 0\right\} \cdot \frac{1}{1 - k}\right\}, \tag{4}$$

  where $p > 0$ and $0 \leqslant k < 1$;

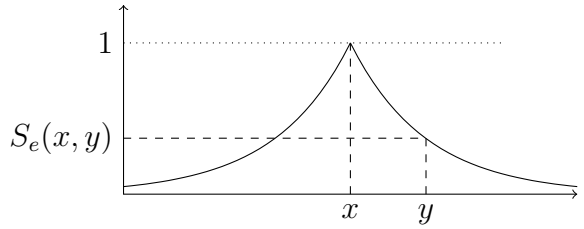


Figure 5: Fuzzy $\star$-similarity relation S_e for $p = 1$ [29].

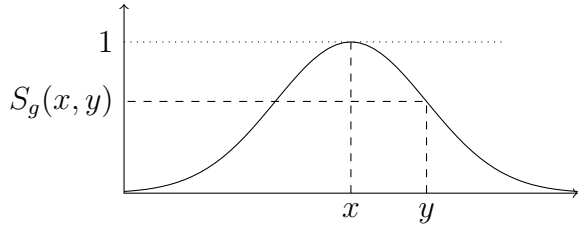


Figure 6: Fuzzy $\star$-similarity relation S_g for $p = 1$ [29].

- for exponential fuzzy sets (Fig. 5)

$$S_e\left(x, y\right) = \exp\left(-\frac{|x - y|}{p}\right), \tag{5}$$

  where $p > 0$;

- for Gaussian fuzzy sets (Fig. 6)

$$S_g\left(x, y\right) = \exp\left(-\frac{(x - y)^2}{2p^2}\right), \tag{6}$$

  where $p > 0$.

Having defined the $\star$-similarity relation the extensional fuzzy number x_S is defined as

$$\forall_{y \in \mathbb{R}} \qquad x_S(y) = S\left(x, y\right). \tag{7}$$

Thus x defines the core of the extensional fuzzy number and the parameter p is responsible for its uzziness. So $x_{S(p)}$ fully defines a extensional fuzzy number.

A one-argument arithmetical operation $\mathbf{op}$ on extensional fuzzy numbers based on a $\star$-similarity relation S is defined as [29]:

$$\mathbf{op}(x_S) = \mathbf{op}(x)_S, \tag{8}$$

for example

$$-\left(x_S\right) = \left(-x\right)_S \tag{9}$$
$$\left(x_S\right)^{-1} = \left(1/x\right)_S, \qquad \text{for } x_S \neq 0_S \tag{10}$$

A two-argument arithmetical operation **op** on extensional fuzzy numbers is defined as [29]:

$$x_{S(p_x)} \,\mathbf{op}\, y_{S(p_y)} = \left(x \,\mathbf{op}\, y\right)_{\max\{p_x,p_y\}}, \tag{11}$$

for example

$$x_{S(p_x)} + y_{S(p_y)} = \left(x + y\right)_{S(\max(p_x,p_y))} \tag{12}$$
$$x_{S(p_x)} \cdot y_{S(p_y)} = \left(x \cdot y\right)_{S(\max(p_x,p_y))} \tag{13}$$

The above formulae make the arithmetic with extensional fuzzy numbers very efficient computationally.

For extensional fuzzy numbers, the equality operator returns a fuzzy value from the interval $[0, 1]$ that is defined as a $\star$-similarity relation S:

$$\left(x_{S(p_x)} =_S y_{S(p_y)}\right) = S_{(\max\{p_x,p_y\})}(x, y). \tag{14}$$

The fuzzy operator $<_S$ is defined as:

$$\left(x_{S(p_x)} <_S y_{S(p_y)}\right) = \begin{cases} 1 - \left(x_{S(p_x)} =_S y_{S(p_y)}\right) & : x \leq y \\ 0 & : x > y \end{cases}. \tag{15}$$

[Nie wiem, czy opisywać pozostałe operacje logiczne, tylko te dwie są wykorzystywane] [Na razie nie :-)]

3 Granular agglomerative hierarchical clustering

The hierarchical clustering algorithms have in general a high time complexity – in the case of the general algorithm up to $O(X^3)$, where X is the number of data points. This problem can be mitigated with granular computation. Instead of clustering the original data, first the data is granulated, and then the hierarchical clustering is applied to the granules. In our approach, the granules are represented as extensional fuzzy numbers. [https://github.com/ZofiaLor/magister-granule]

In the first step (line 2 in Alg. 1), the original data $\mathbb{X}$ is clustered with the fuzzy c-means algorithm FCM. FCM returns the matrix of centres $\mathbf{V}$ of G clusters and

Algorithm 1 GRAGGLO – granular agglomerative hierarchical clustering

Require: $\mathbb{X}$: input data
Require: G: number of granules
Require: C: expected number of clusters
Require: r: type of the $\star$-relation
Require: ξ: threshold for the fuzzy minimum operation

procedure GRAGGLO(zX, G, r, ξ)
  $\mathbf{V}, \mathbf{U} \leftarrow$ FCM($\mathbb{X}, G$) ▷ clusterisation of the input data, $\mathbf{V}$: centres of clusters, $\mathbf{U}$: memberships of data items to clusters, $O(XG)$
  $\mathbb{G} \leftarrow$ CLUSTERSTOGRANULES($\mathbf{V}, \mathbf{U}, r$) ▷ $O(XG)$
  for each granule g_1 in $\mathbb{G}$ **do**
    for each granule $g_2 \neq g_1$ in $\mathbb{G}$ **do**
      $\mathbf{D}[g_1, g_2] \leftarrow$ FUZZYDISTANCE(g_1, g_2)
    end for
  end for
  $n \leftarrow G$ ▷ each granule is a separate cluster
  while $n > C$ **do** ▷ $O(G)$
    $i, j \leftarrow$ FINDINDICESOFTHECLOSESTGRANULES($\mathbf{D}, \mathbf{p}, r, \xi$) ▷ $O(G^2)$
    $\mathbf{r} \leftarrow [\,]$ ▷ for the merged cluster, create a new row $\mathbf{r}$…
    $\mathbf{c} \leftarrow [\,]$ ▷ … and a new column $\mathbf{c}$ for the distance matrix $\mathbf{D}$
    for all granules g in $\mathbb{G}$ **do** ▷ merge the found clusters into one
      $\mathbf{r}[g] = \mathbf{c}[g] =$ FUZZYAGGREGATION($\mathbf{D}[i, g], \mathbf{D}[j, g], r, \xi$)
    end for
    remove the i-th and j-th rows and columns from the matrix $\mathbf{D}$
    add the new row $\mathbf{r}$ and column $\mathbf{c}$ to the distance matrix $\mathbf{D}$
    $n \leftarrow n - 1$ ▷ update the number of clusters
  end while
end procedure

the matrix $\mathbf{U}$ of memberships of data points to clusters. The obtained results are then transformed into granules with the ClustersToGranules procedure (line 3 in Alg. 1). Granules are represented with extensional fuzzy numbers, which are defined by the following parameters: the centre of the granule v_{cd}, which is equal to the centre of the cluster obtained from FCM, and the fuzziness p_{cd}, which is calculated based on the memberships $\mathbf{U}$ of data points to clusters with the following formula:

$$p_{cd} = \sqrt{\frac{\sum_{i=1}^{X} \mu_{ci}^{m} (x_{id} - v_{cd})^2}{\sum_{i=1}^{X} \mu_{ci}^{m}}}. \tag{16}$$

Then (lines 4–8) two nested loops are responsible for elaboration of fuzzy distance between granules. The distances are stored in the distance matrix $\mathbf{D}$. The crucial part of the algorithm is the while loop (lines 10–20), which implements the agglomerative hierarchical clustering of granules. Each iteration of the loop merges the two closest granules into one, and updates the distance matrix $\mathbf{D}$ according to the selected type of aggregation method in the FuzzyAggregation procedure (line 15). In the granular single-linkage method, the fuzzy minimum operation is used, while in the granular complete-linkage method, the fuzzy maximum operation is used. Thus, each iteration reduces the number of granules by one (line 19), until the expected number of clusters C is reached.

The granulation of the input data $\mathbb{X}$ with the FCM algorithm and tranformation of clusters into granules has a time complexity of $O(XG)$, where X is the number of data points and G number of granules. The most time-consuming part of the algorithm is the while loop, which has to be executed $G - C$ times. In each iteration, the algorithm has to find the two closest granules in the distance matrix $\mathbf{D}$, which has a size of $G \times G$. This makes the time complexity of the while loop $O(G^3)$. Finally, the overall time complexity of the GRAgglo algorithm is $O(XG + G^3)$, which is significantly lower than the time complexity of the general agglomerative hierarchical clustering algorithm $O(X^3)$, because $G \ll X$.

4 Experiments

The experiments aim at evaluating the performance of the proposed granular agglomerative hierarchical clustering algorithm GRAgglo in comparison to the general agglomerative hierarchical clustering algorithm in the perspective of the time complexity and the quality of clustering results.

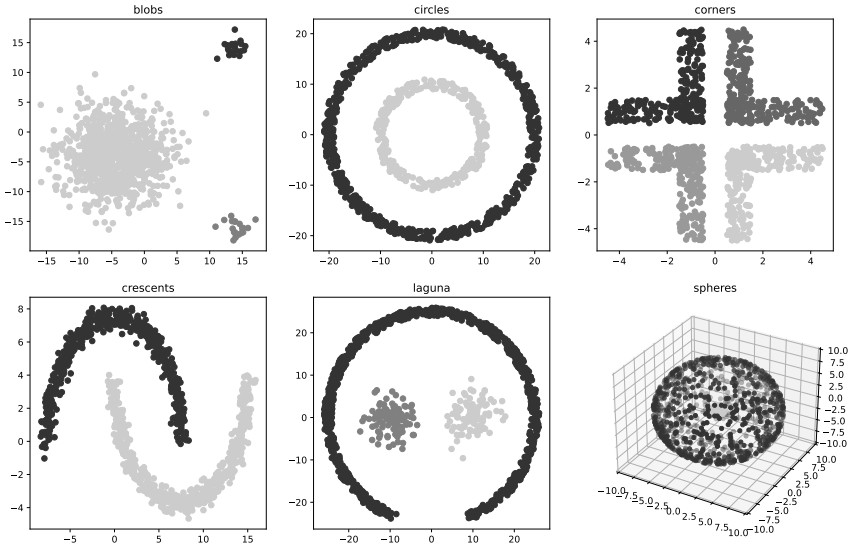


Figure 7: Visualization of synthetic datasets of size 1000. Points of the same color belong to the same target cluster.

4.1 Datasets

Two groups of datasets are used in the experiments. The first one is synthetic labelled data in six shapes ('blobs', 'circles', 'corners', 'crescents', 'laguna'and 'spheres') and eight sizes (1000, 2000, 5000, 10000, 20000, 30000, 40000 and 50000 elements [¿Czy tutaj są dobre wartości?]. The data is visualized in Fig. 7. The second type are the real-life unlabelled 2D data 'brightkite'[1] (with 4 491 143 data items) and 'gowalla'[2] (with 6 442 890 data items) [4]. In these two datesets, only the latitude and longitude columns are used for the experiments. The data is visualized in Fig. 8.

[1] `https://snap.stanford.edu/data/loc-brightkite.html`
[2] `https://snap.stanford.edu/data/loc-gowalla.html`

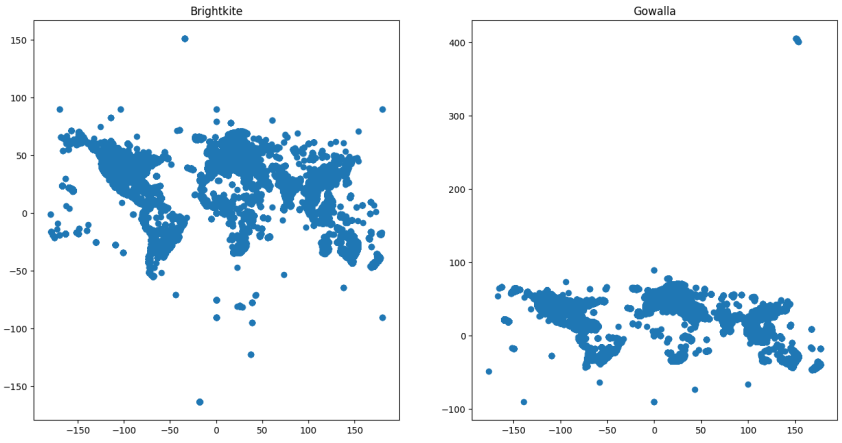


Figure 8: Visualization of ‘brightkite’ and ‘gowalla’ datasets.

Table 1: Comparison of the average execution times (in [ms]) of the agglomerative clustering and the GRAGGLO for different data sizes and granule numbers.

	agglomerative	GRAGGLO, number of granules		
data size	clustering	50	100	200
500	256.15	79.31	166.93	527.55
1000	1092.4	131.34	300.99	944.47
1500	2725.91	189.25	564.14	1269.19
2000	7125.25	280.46	664.66	1546.45
2500	14676.58	331.97	821.36	1848.87
3000	24289.29	474.25	1000.07	2197.36
3500	36747.48	554.55	1122.44	2546.56
4000	52439.32	629.62	1310.48	2901.85
4500	71895.08	696.23	1460.06	3233.01
5000	95497.87	778.06	1590.62	3540.35

4.2 Clustering time

The clustering time is measured for both the general agglomerative hierarchical clustering algorithm and the proposed granular agglomerative hierarchical clustering algorithm GRAGGLO. The experiments were run for various number of input data points and various number of granules.

All experiments confirm the linear time complexity of GRAGGLO algorithm. Fig. 9 and Table 1 compare the execution time of the general agglomerative hierarchical clustering algorithm and the GRAGGLO algorithm for different numbers of data points and granules. The results show that the GRAGGLO algorithm is significantly faster.

Figure 10 and Table 2 compare GRAGGLO with the optimised implementation `AgglomerativeClustering` in the `scikit-learn` library. For more than 40000 data points, GRAGGLO algorithm is faster than `AgglomerativeClustering` even with 200 granules.

An interesting issue needs a short comment. The memory complexity of FCM must be taken into account: the algorithm allocates a membership array of floating-point numbers with size $G \times X$. Figure 11 and Tab. 3 show that for a combination of large numbers of data points and granules, the execution time may increase significantly (200 granules, above 3 million data points case) or FCM might fail to allocate the array due to lack of memory (100 and 200 granules, 6 million data points case).

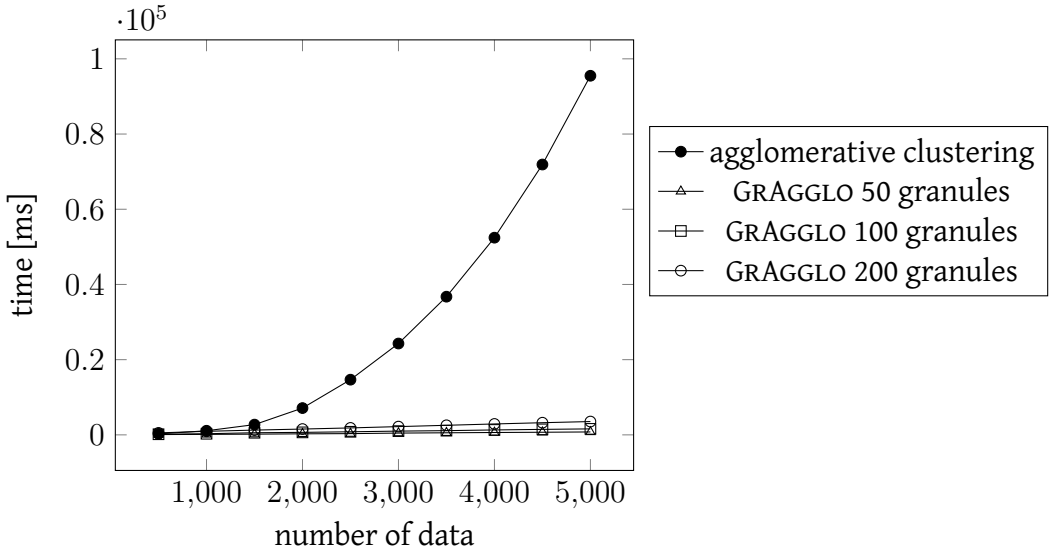


Figure 9: Comparison of the average execution times (in [ms]) of the agglomerative clustering and the GRAGGLO for different data sizes and granule numbers.

Table 2: Comparison of average execution times (in [ms]) of the `scikit-learn` library's strict and proposed fuzzy agglomerative hierarchical clustering algorithms for different data and granule numbers.

	agglomerative	GRAGGLO, number of granules		
data size	clustering	50	100	200
1000	27.68	140.37	278.82	922.37
2000	39.44	252.88	662.75	1488.29
5000	312.70	774.75	1535.61	3344.13
10000	1425.18	1465.33	3017.63	6566.77
20000	6434.67	2925.11	6101.59	13076.91
30000	15171.27	4659.39	9518.74	20115.51
40000	29034.08	6226.38	12677.87	26848.27
50000	47815.40	7811.42	15775.42	33288.67

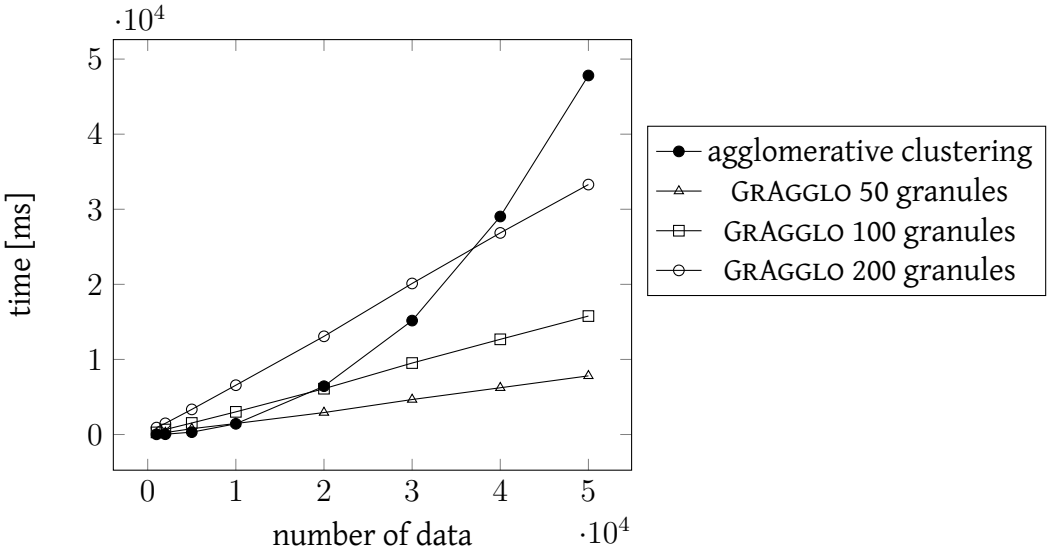


Figure 10: Comparison of average execution times of the `scikit-learn` library's strict and proposed fuzzy agglomerative hierarchical clustering algorithms for different data and granule numbers.

Table 3: Comparison of execution times (in [s]) of fuzzy agglomerative hierarchical clustering algorithm GRAGGLO for different data and granule numbers, for large data numbers.

	number of granules		
data size	50	100	200
50000	7.72	15.84	33.23
100000	15.53	31.72	67.03
500000	79.73	177.07	348.11
1000000	158.69	322.65	675.32
2000000	300.08	641.60	1348.01
3000000	468.20	967.35	2126.28
4000000	628.20	1328.02	5400.40
5000000	799.02	1696.94	8650.32
6000000	964.80	–	–

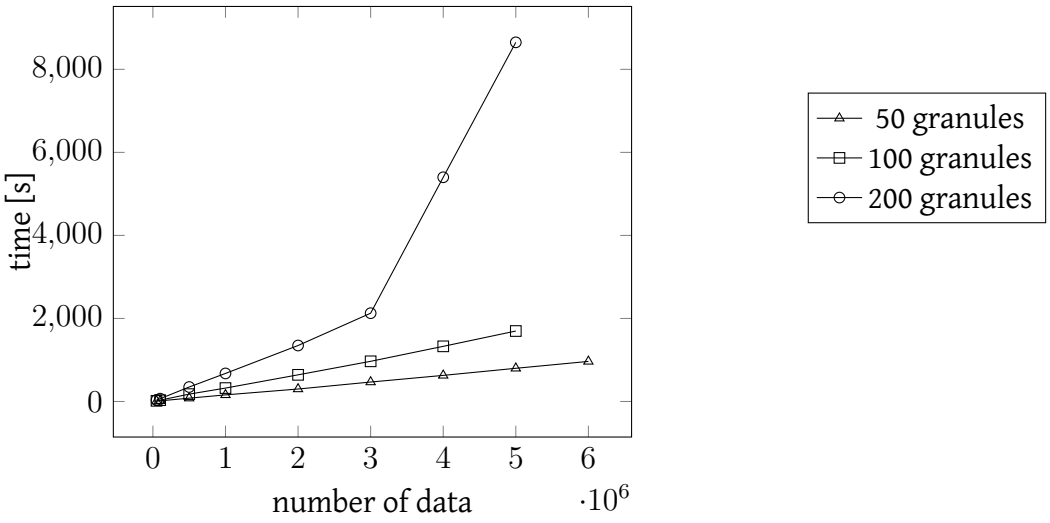


Figure 11: Comparison of execution times of fuzzy agglomerative hierarchical clustering algorithm GRAGGLO for different data and granule numbers, for large data numbers.

4.3 Quality of clustering

The aim of the GRAGGLO algorithm is to reduce the time complexity of the agglomerative hierarchical clustering algorithm, while maintaining the quality of clustering results. Our approach is expected to elaborate the same or similar clustering results as the general agglomerative hierarchical clustering algorithm, but in a shorter time. GRAGGLO clusters fuzzy granules instead of the original crips data. The assignment of original data points to clusters is based on the memberships of the data items to fuzzy granules and the memberships of fuzzy granules to clusters. The procedure is presented in Alg. 2. The procedure takes as input the original data points $\mathbb{X}$, the granules $\mathbb{G}$ and their labels $\mathbf{e}_g$. For each data item $\mathbf{x}$ (line 4, the procedure calculates its membership to each granule (line 7). The data item is assigned to the granule with the highest membership value (line 14). In case of granules based on triangular extensional fuzzy numbers, it is possible that the membership of a data point to all granules is equal to 0. This is due to the finite support of triangular extensional fuzzy numbers. The procedure returns the array of labels $\mathbf{e}_x$ with labels for each data point (line 24).

[To jest bardziej skomplikowany sposób, który był omawiany w pracy magisterskiej. Poza tym, że mogły pojawić się szumy, nie wprowadził on żadnych nowych informacji. Wszystkie tabele z wartościami wyników, które zostały w artykule, dotyczą prostszej metody, opisanej w następującym akapicie:]

Each data item $\mathbf{x}_i$ is assigned to the granule g_j, such that the value of $\mu_{ij} = \max \boldsymbol{\mu}_i$.

Two arrays of labels are compared: the first one is the array of labels $\mathbf{e}_x$ elaborated by the GRAGGLO algorithm or the general agglomerative hierarchical clustering algorithm, and the second one is the array of expected labels $\mathbf{e}_x^{\mathrm{a}}$ [To oznaczenie trzeba będzie zmienić na coś sensowniejszego, ale na razie to tak zostawiam.]. Thus we get the multiclass classification problem and we use the confusion matrix $\mathbf{L}$ to measure the quality of clustering of either algorithm. The expected labels are considered as the ground truth, while the labels elaborated by GRAGGLO or the crisp agglomerative hierarchical clustering algorithm are considered as predicted labels.

The total accuracy A is elaborated from the confusion matrix $\mathbf{L}$ as follows.

$$A = \frac{\sum_{i=0}^{C} l_{ii}}{\sum_{i=0}^{C} \sum_{j=0}^{C} l_{ij}} = \frac{\sum_{i=0}^{C} l_{ii}}{X}, \tag{17}$$

where X is the number of data points. The recall R and precision P are first calculated separately for each cluster c:

$$R_c = \frac{l_{cc}}{\sum_{i=0}^{C} l_{ci}}, \tag{18}$$

$$P_c = \frac{l_{cc}}{\sum_{i=0}^{C} l_{ic}}. \tag{19}$$

The the recall R and precision P for the whole clustering are calculated as the arithmetic mean of the partial metrics:

$$R = \frac{\sum_{i=0}^{C} R_i}{C}, \tag{20}$$

where C is the number of clusters, and

$$P = \frac{\sum_{i=0}^{C} P_i}{C}. \tag{21}$$

[Do wyrzucenia, jeśli nie opisujemy metody z alg. 2] In case if the noise set $\mathbb{N}$ is not empty, two approaches are possible to calculate the metrics:

1. The noise is not taken into account, and the metrics are calculated only for the data points that are not noise.

2. The noise is considered as misclassified data. To correct the value of the metric for noise, it should be multiplied by the percentage of non-noise points in relation to the total data.

The noise set is not empty if the fuzzy relation produces a fuzzy set with a finite support. In our research, this is the case of triangular extensional fuzzy numbers (Fig. 3).

Tab. 4-9 present the maximal accuracy, recall, and precision for the single and complete linkage, respectively for all combinations of ξ values and granule shapes, averaged for each original data size. The values are 1.0000 for all almost all datasets, data sizes, and numbers of granules. Only for some cases the values are slightly lower and for all cases except 'blobs' and 200 granules they are still higher than the crisp algorithm's results. 0.9696

Fig. 12-15 show the impact of the ξ parameter's value and the number of granules on the results. For SINGLELINKAGE, the lower the value of ξ the better. The results are usually better for larger numbers of granules though it depends on data shape and distribution. Fig. 14 shows an incorrect clustering for 100 granules while both the lower (50 granules) and higher (200 granules) with the same $\xi = 0.9$ value are clustered according to expectations. [Zależność odwróconą mamy też dla 'blobs' – może warto jednak pokazać ten wykres?] [Tak, dajmy to.]

The lower results of clusterization of the 'laguna' dataset are explained in Fig. 16. For smaller datasets, all clusters are clearly separated. For larger datasets the middle two clusters are joined, the datapoints in the middle of that joining

Table 4: Maximal accuracy of hierarchical clustering of data with the strict SINGLELINKAGE algorithm

	accuracy			
	crisp data	granular data		
dataset		50 granules	100 granules	200 granules
'blobs'	0.9899	0.9999	0.9939	0.9696
'circles'	1.0000	1.0000	1.0000	1.0000
'corners'	1.0000	1.0000	1.0000	1.0000
'crescents'	1.0000	1.0000	1.0000	1.0000
'laguna'	0.9125	0.9997	0.9998	0.9999
'spheres'	1.0000	1.0000	1.0000	1.0000

might be classified as the other cluster than expected. Notably, GRAGGLO performs better than the crisp SINGLELINKAGE algorithm, which tends to merge the two middle clusters into one.

The impact of fuzzy relation type is the least pronounced. For each dataset, 100% (or almost 100% in 'blobs' and 'laguna' cases) accuracy can be achieved with any of the 3 relation types. However, there are differences when every parameter combination is taken into account: the total average accuracy is then highest for triangular relation and lowest for exponential relation. [Czy to jest związane z tym, że trójkątna relacja może generować szum?] [Tabela? Wykres?]

Clustering of the 'brightkite' dataset is shown in Fig. 17. Since the dataset is unlabelled, the results can be evaluated visually. Clustering with the strict algorithm produces one big cluster consisting of most of the data and five small clusters representing singular points. Clustering with GRAGGLO results in groups mostly split by continent, concentrated in areas of higher density of datapoints. It is considerably closer to the expected result compared to the strict algorithm.

5 Summary

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Table 5: Maximal recall of hierarchical clustering of data with the strict SINGLE-LINKAGE algorithm

	recall			
	crisp data	granular data		
dataset		50 granules	100 granules	200 granules
'blobs'	0.8333	1.0000	0.9571	0.7487
'circles'	1.0000	1.0000	1.0000	1.0000
'corners'	1.0000	1.0000	1.0000	1.0000
'crescents'	1.0000	1.0000	1.0000	1.0000
'laguna'	0.7084	0.9990	0.9995	0.9997
'spheres'	1.0000	1.0000	1.0000	1.0000

Table 6: Maximal precision of hierarchical clustering of data with the strict SINGLELINKAGE algorithm

	precision			
	crisp data	granular data		
dataset		50 granules	100 granules	200 granules
'blobs'	0.8295	0.9990	0.9392	0.7282
'circles'	1.0000	1.0000	1.0000	1.0000
'corners'	1.0000	1.0000	1.0000	1.0000
'crescents'	1.0000	1.0000	1.0000	1.0000
'laguna'	0.8375	0.9996	0.9998	0.9998
'spheres'	1.0000	1.0000	1.0000	1.0000

Table 7: Maximal accuracy of hierarchical clustering of data with the strict CompleteLinkage algorithm

	accuracy			
	crisp data	granular data		
dataset		50 granules	100 granules	200 granules
'blobs'	0.7261	0.9999	0.8956	0.7237
'circles'	0.5620	0.5533	0.5330	0.5627
'corners'	0.5265	0.4818	0.5053	0.5130
'crescents'	0.7986	0.8012	0.8032	0.7822
'laguna'	0.4508	0.4875	0.4653	0.4664
'spheres'	0.5852	0.5712	0.5650	0.6615

Table 8: Maximal recall of hierarchical clustering of data with the strict CompleteLinkage algorithm

	recall			
	crisp data	granular data		
dataset		50 granules	100 granules	200 granules
'blobs'	0.7825	1.0000	0.8822	0.9041
'circles'	0.4900	0.4940	0.4876	0.5071
'corners'	0.5265	0.4818	0.5053	0.5130
'crescents'	0.7986	0.8012	0.8033	0.7822
'laguna'	0.5447	0.6026	0.4856	0.5072
'spheres'	0.5473	0.3729	0.4805	0.5897

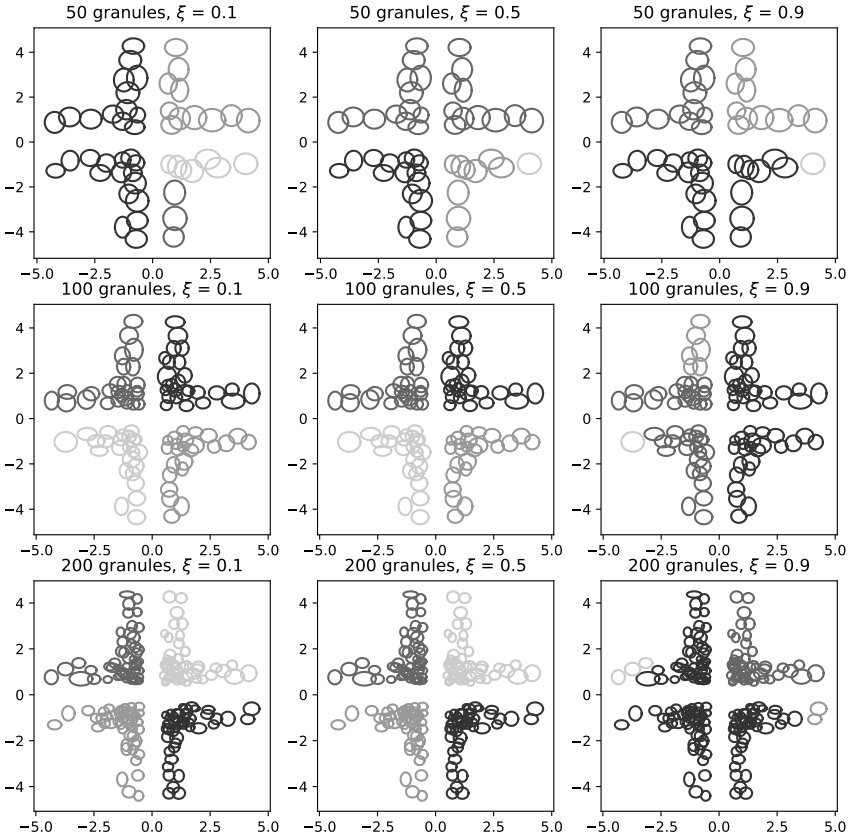


Figure 12: Results of clustering GRAGGLO for the 'corners' dataset of size 1000 and gaussian relation between fuzzy numbers, with SINGLELINKAGE connectivity.

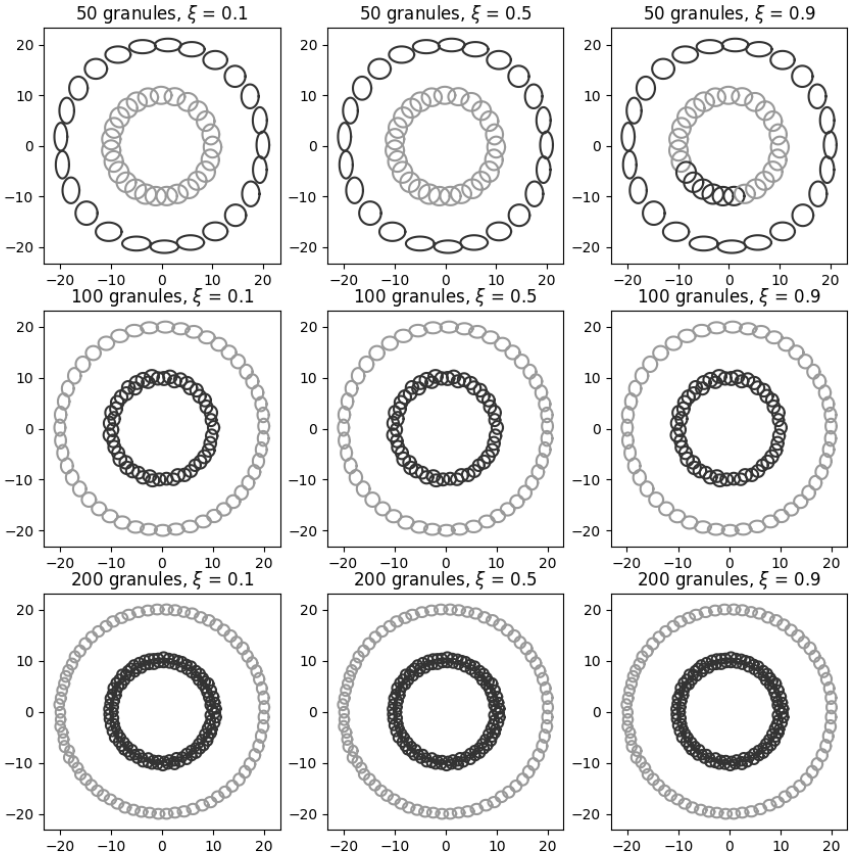


Figure 13: Results of clustering GRAGGLO for the 'circles' dataset of size $50\,000$ and exponential relation between fuzzy numbers, with SINGLELINKAGE connectivity.

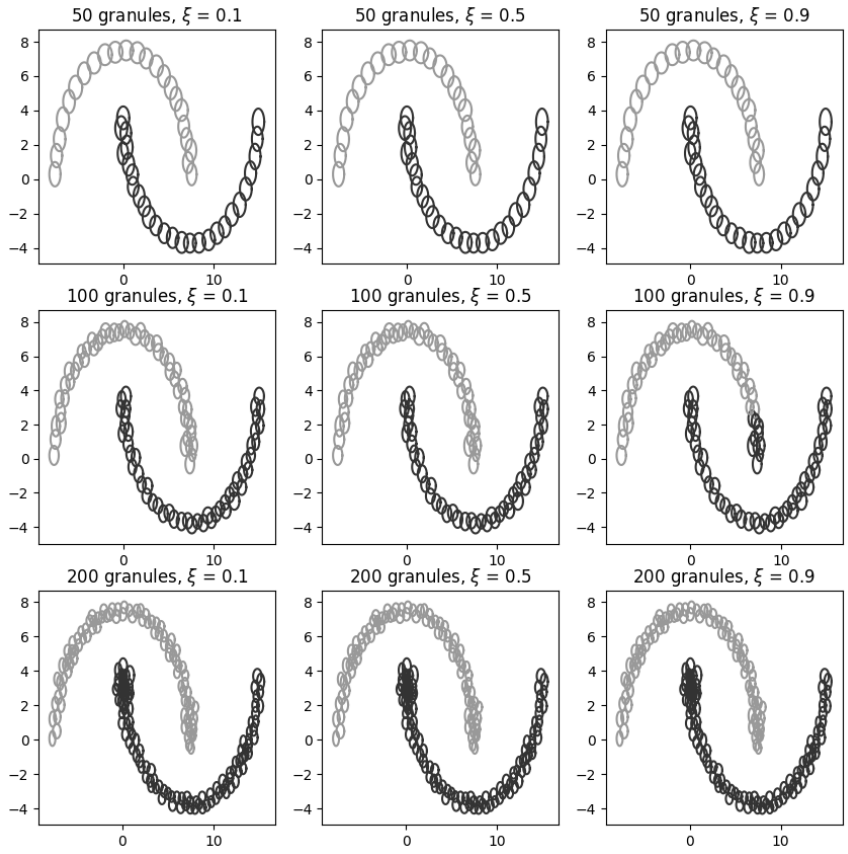


Figure 14: Results of clustering GRAGGLO for the 'crescents' dataset of size 5000 and exponential relation between fuzzy numbers, with SINGLELINKAGE connectivity.

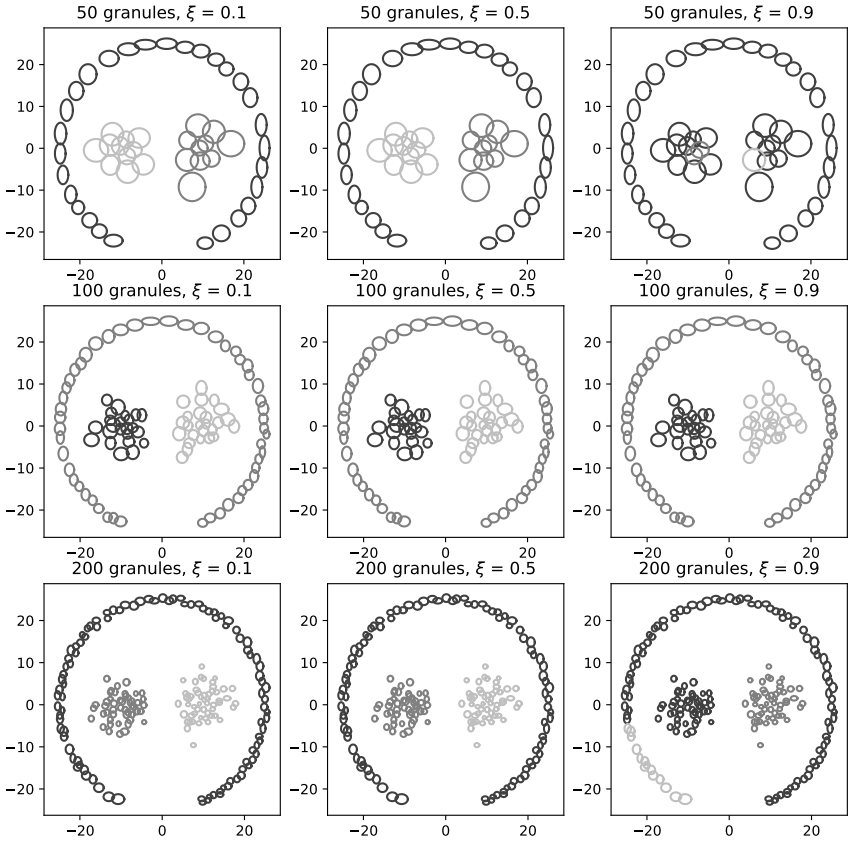
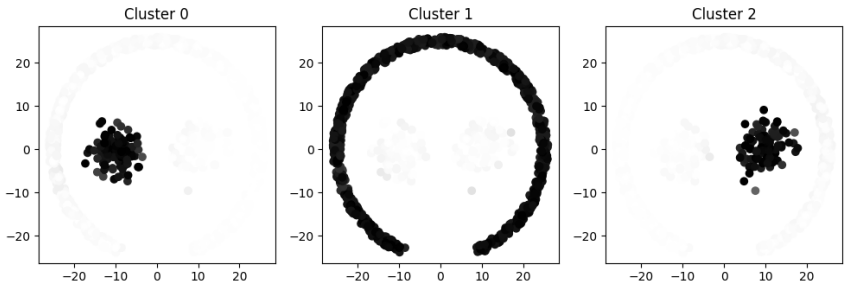


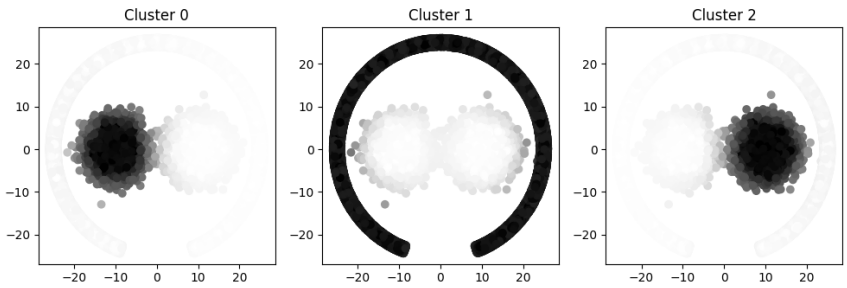
Figure 15: Results of clustering GRAGGLO for the 'laguna' dataset of size 1000 and exponential relation between fuzzy numbers, with SINGLELINKAGE connectivity.

Table 9: Maximal precision of hierarchical clustering of data with the strict CompleteLinkage algorithm

	precision			
	crisp data	granular data		
dataset		50 granules	100 granules	200 granules
‘blobs’	0.6054	0.9990	0.8753	0.6591
‘circles’	0.4881	0.4948	0.4880	0.5052
‘corners’	0.4956	0.4779	0.4631	0.4828
‘crescents’	0.8372	0.8072	0.8285	0.8093
‘laguna’	0.3984	0.4302	0.3561	0.3804
‘spheres’	0.5143	0.4480	0.4891	0.5324

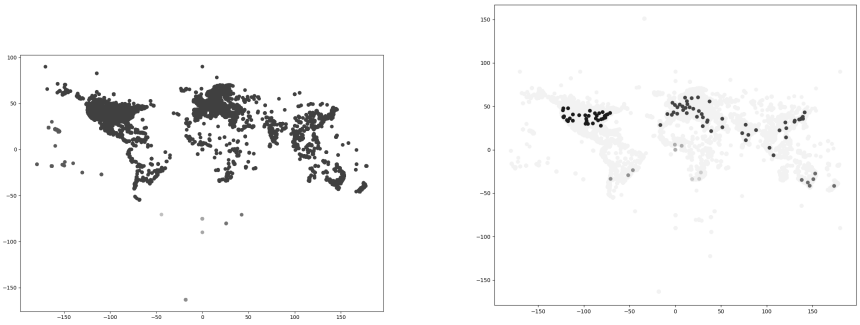


(a) 1000 points



(b) 40 000 data points

Figure 16: Membership of points to clusters for the ‘laguna’ dataset.



(a) Clustering of a random sample of $500\,000$ points with the strict algorithm.

(b) Clustering of the entire dataset with the GRAGGLO algorithm with 100 granules, triangular fuzzy relation, $\xi = 0.1$.

Figure 17: Results of hierarchical clustering with the crisp agglomerative clustering algorithm and the GRAGGLO algorithm for the 'brightkite' dataset.

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Algorithm 2 Elaboration of the membership of a data point to a granule based on the $\star$-similarity relation S

Require: $\mathbb{X}$: original data points
Require: $\mathbb{G}$: granules elaborated by GRAGGLO
Require: $\mathbf{e}_g$: array of label elaborated for granules
Require: r: relation

procedure CALCULATESIMILARITYMEMBERSHIP($\mathbb{X}, \mathbb{G}, \mathbf{e}_g, r$)
  $\mathbf{e}_x \leftarrow$ array 1..X of $\varnothing$ ▷ array of labels for each data point
  $\mathbb{N} \leftarrow 0$ ▷ noise: set of data not belonging to any granule
  for $\mathbf{x} \in \mathbb{X}$ **do**
    $e \leftarrow 0$
    $\mu_{max} \leftarrow 0$
    for $g \in \mathbb{G}$ **do**
      $\mu \leftarrow 1$ ▷ initial membership of the data point to the granule
      for $d \in \mathbb{D}$ **do** ▷ for each attribute d of data point $\mathbf{x}$ and granule g
        $\mu \leftarrow \mu \star (g[d] =_S \mathbf{x}[d])$ ▷ $\star$ t-norm
      end for
      if $\mu > \mu_{max}$ **then**
        $\mu_{max} \leftarrow \mu$
        $e \leftarrow \mathbf{e}_g[g]$ ▷ label of the granule g
      end if
    end for
    if $\mu_{max} > 0$ **then**
      $\mathbf{e}_x \leftarrow e$ ▷ assign the label of the data point
    else ▷ data item does not belong to any granule
      $\mathbf{e}_x \leftarrow \varnothing$
      $\mathbb{N} \leftarrow \mathbb{N} \cup \mathbf{x}$
    end if
  end for
  return $\mathbf{e}_x$ ▷ vector of labels for each data point
end procedure